

1) $D \in \{0,1\}$ detection

$B \in \{0,1\}$ balloon

a) - want $P(D=0|B=1)$

- know that $\sum_{d \in \{0,1\}} P(D=d|B=1) = 1$

$$\therefore P(D=0|B=1) = 1 - P(D=1|B=1) = 1 - 0.63 = \boxed{0.37}$$

b) - want $P(B=1|D=0)$

- know $P(D|B)$

- Bayes rule

$$P(B|D) = \frac{P(D|B)P(B)}{P(D)}$$

- If we know $P(B)$, we can calculate $P(D)$ with $P(D) = \sum_b P(D|B=b)P(B=b)$.

$\therefore$ We need to know the marginal distribution of a balloon passing over, $P(B)$ to calculate $P(B|D)$

c) redefine $B \in \{0, 1, 2\}$
no balloon weather surveillance

B	P(B)
0	0.34
1	0.47
2	0.19

want $P(B=2 | D=1)$

$$P(B|D) = \frac{P(D|B) P(B)}{P(D)}$$

$$P(D=1) = \sum_b P(D=1|B=b) P(B=b)$$

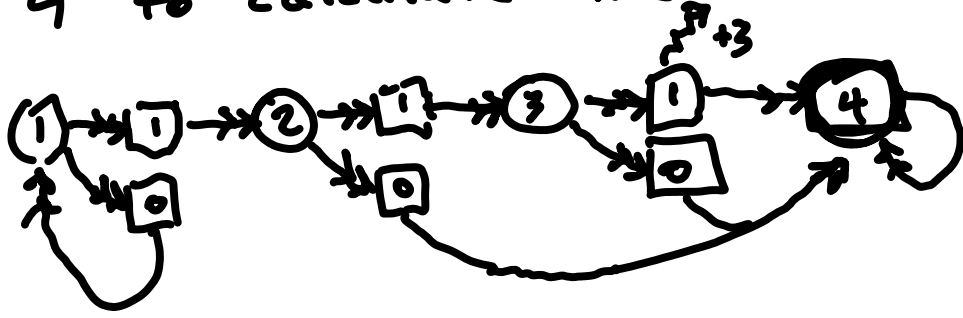
$$= 0.07 \cdot 0.34 + 0.63 \cdot 0.47 + 0.63 \cdot 0.19$$

$$= 0.4396$$

$$P(B=2|D=1) = \frac{P(D=1|B=2) P(B=2)}{P(D=1)}$$

$$= \frac{0.63 \cdot 0.19}{0.4396} = \boxed{0.272}$$

2) State 4 is a terminal state since there will be no transitions away from it and it has no reward. We can work backwards from 4 to calculate the value.



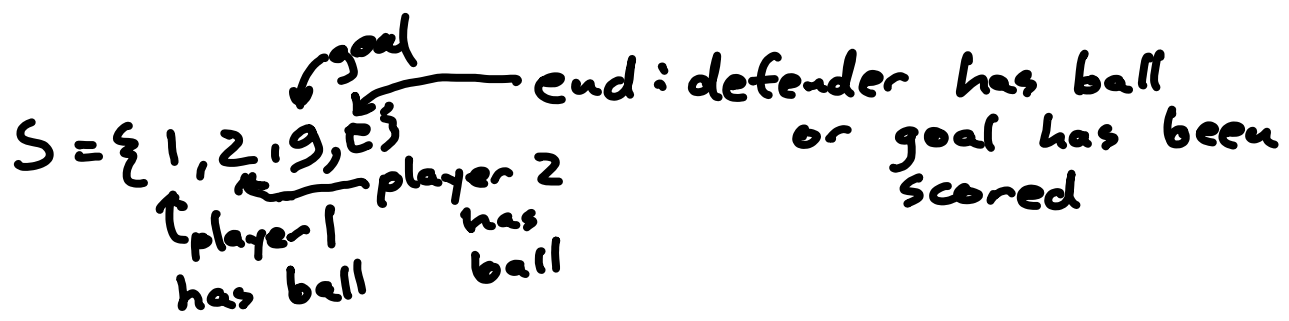
s	$V^*(s)$	$\pi^*(s)$	a	$Q^*(s,a)$
4	0	0	0	$0 + \gamma V^*(4)$
3	3	1	1	$3 + \gamma V^*(4) = 3$
-	-	-	0	$0 + \gamma V^*(4) = 0$
2	2.7	1	1	$0 + \gamma V^*(3) = 0.9 \cdot 3 = 2.7$
-	-	-	0	$0 + \gamma V^*(1) = 0$
1	2.43	1	1	$0 + \gamma V^*(2) = 0.9 \cdot 2.7 = 2.43$

optimal value

optimal policy

To prove that this is 0, let π_1 be a policy where $\pi_1(i) = 0$ and $\pi_1(s)$ is the previously-calculated policy for all other states. Then we have $V^{\pi_1}(1) = 0 + \gamma V^{\pi_1}(1) \therefore V^{\pi_1}(1) = 0$.

3] (there are many correct answers) 10:00



$$A = \{\text{pass}, \text{shoot}\}$$

$$R(s, a) = \begin{cases} 1 & \text{if } s = g \\ 0 & \text{o.w.} \end{cases}$$

$$T^{\text{pass}} = \begin{matrix} & \begin{matrix} 1 & 2 & g & e \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ g \\ e \end{matrix} & \begin{bmatrix} 0 & 0.95 & 0 & 0.15 \\ 0.8 & 0 & 0 & 0.2 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

$$T^{\text{shoot}} = \begin{matrix} & \begin{matrix} 1 & 2 & g & e \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ g \\ e \end{matrix} & \begin{bmatrix} 0 & 0 & 0.35 & 0.65 \\ 0 & 0 & 0.45 & 0.55 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

$\gamma = 0.9$ (because there is a 10% chance of the game ending on every step)

4) Steps:

1. Create transition matrix, T^π , and reward vector R^π for the policy.
2. Evaluate the policy to find the policy value function:

$$V^\pi = (I - \gamma T^\pi)^{-1} R^\pi$$

3. Check if this value function satisfies Bellman's optimality equation

$$V^\pi(s) \stackrel{?}{=} \max_a \left(R(s,a) + \gamma \sum_{s'} T(s'|s,a) V^\pi(s') \right)$$

If this is satisfied for all s , V^π is the optimal Value function and therefore π is an optimal policy.

5) $R(s,a) = -s^2 - a^2$ will have the desired optimal policy. This is because the problem fits the form of a Linear Quadratic Regulator (LQR) problem. Specifically, the reward has the form

$R(s,a) = s^T R_s s + a^T R_a a$ where R_s is negative semi-definite and R_a is negative definite.